

- 6 (a) Fig. 6.1 is a ribbon model of a molecule of haemoglobin.

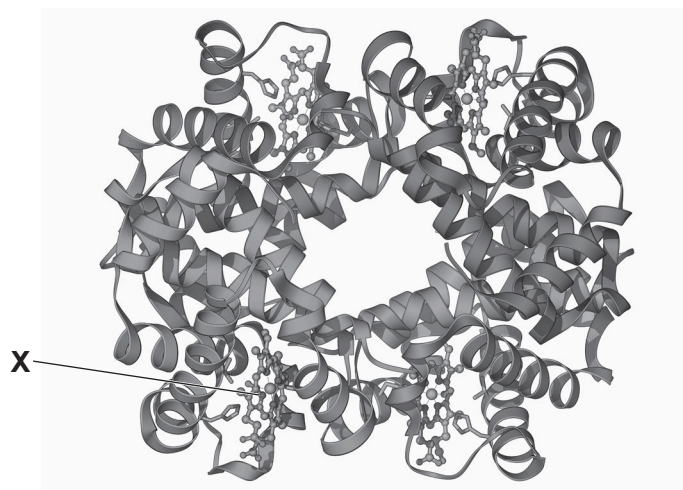


Fig. 6.1

- (i) State the part of the haemoglobin molecule labelled X.

..... [1]

- (ii) State the function of the structure labelled X.

.....
..... [1]

- (iii) Haemoglobin is described as having quaternary structure.

State what is meant by quaternary structure.

.....
..... [1]

- (b) The effect of the partial pressure of oxygen (pO_2) and the effect of the partial pressure of carbon dioxide (pCO_2) on the percentage saturation of haemoglobin was investigated.

A sample of mammalian blood was exposed to a gas mixture that contained increasing pO_2 . In the experiment, the pCO_2 was maintained at 2.7 kPa. The percentage saturation of haemoglobin in the blood sample was determined as the pO_2 increased.

The experiment was repeated with further samples of blood with a pCO_2 maintained at 5.3 kPa and at 10.7 kPa.

The results are shown in Fig. 6.2.

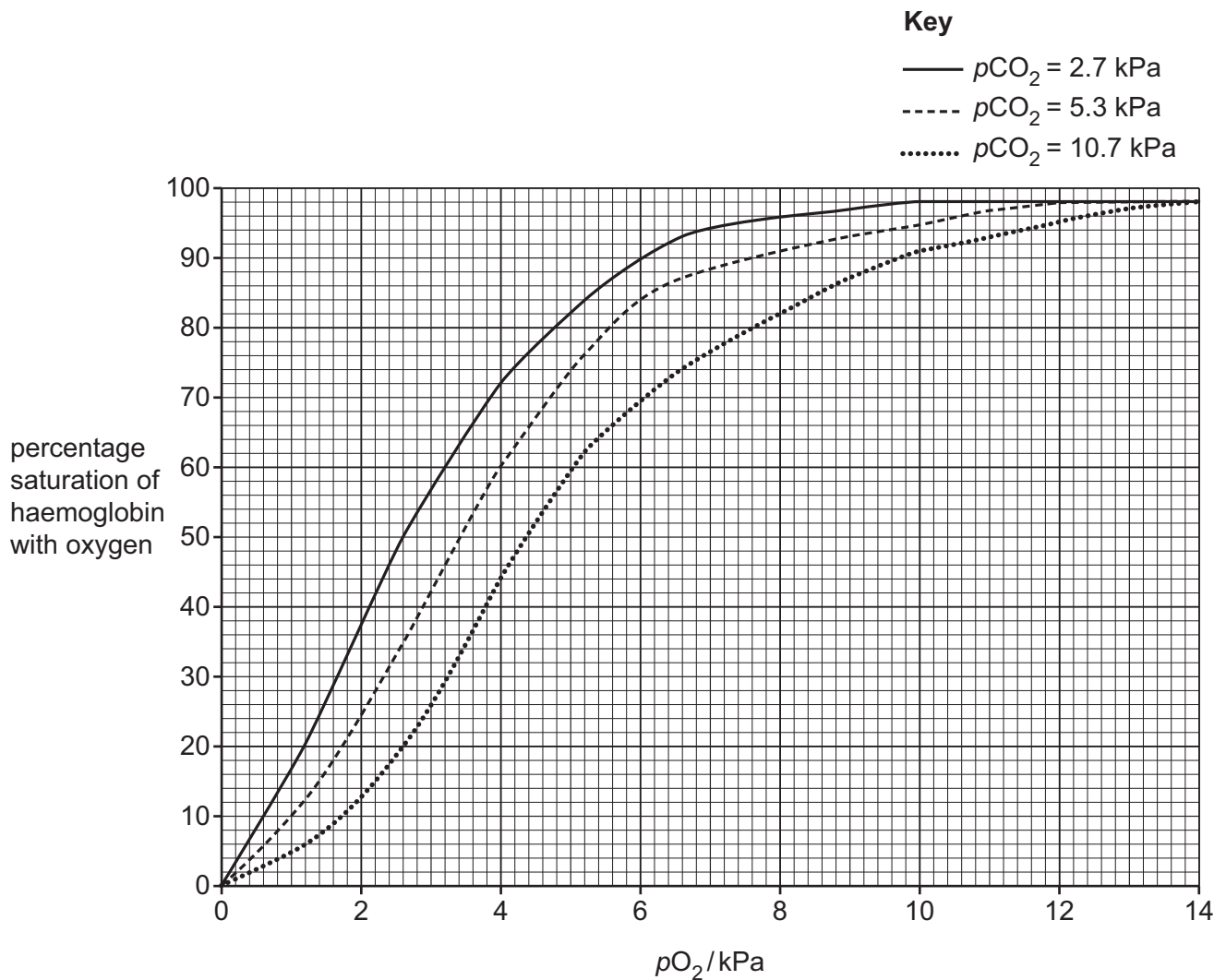


Fig. 6.2

- (i) The $p\text{CO}_2$ of alveolar air is 5.3 kPa.

With reference to Fig. 6.2, state the likely partial pressure of oxygen in the alveoli of the mammal.

..... [1]

- (ii) Suggest the range of partial pressures of oxygen in respiring tissues **and** use Fig. 6.2 to give evidence for your answer.

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..... [2]



- (iii) Use the information in Fig. 6.2 to describe the effect of increasing $p\text{CO}_2$ on the percentage saturation of haemoglobin with oxygen.

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..... [3]

- (iv) State the name given to the effect you have described in part (iii).

..... [1]

- (v) Explain the advantage to the mammal of the effect you described in part (iii).

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..... [2]

[Total: 12]

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